

# Lordphone Wen

Mountain View, CA

[lordphonewen@gmail.com](mailto:lordphonewen@gmail.com) · (608) 217-8920 · [linkedin.com/in/lordphone-wen](https://www.linkedin.com/in/lordphone-wen) · [github.com/lordphone](https://github.com/lordphone)

## EDUCATION

### Carnegie Mellon University

Expected May 2027

*M.S. in Software Engineering | GPA: 4.0 / 4.0*

Selected Coursework: Distributed Systems, Data Science for Software Engineering, Machine Learning

### University of Wisconsin–Madison

May 2024

*B.S. in Computer Science & B.S. in Computer Engineering | GPA: 3.72 / 4.0*

Selected Coursework: Operating Systems, Algorithms, Computer Architecture

## WORK EXPERIENCE

### NewsBreak

May – August 2026

*Backend Engineer Intern*

*Mountain View, CA*

- Built SportsBreak's Python FastAPI / MongoDB backend from its first commit to 89 endpoints and 26 collections, with ~1,450 fully offline tests behind a ruff / pyright / pytest gate; authored 344 PRs.
- Stood up production Kubernetes (AWS EKS, Helm, Jenkins) on day 3 with 7 CronJobs; fixed a 5-day silent CronJob outage (ImagePullBackOff + Forbid concurrency) with paired job and scheduler deadlines.
- Redesigned a cache that could not hit (13.2MB keyed per user) into a shared 10.9MB catalog cache, 78× smaller, sized for a 512MiB pod; replaced a 25–30 request client fan-out with one backend call.
- Designed an ingest → embed → LLM-score → dedupe → push pipeline (~1,200 LLM calls/day); calibrated on 4,100+ shadow decisions, it cut filler 34% → 4% and held its 2/day cap over 11,745 user-days.
- Built hybrid retrieval (Mongo text + vector, RRF) that degrades to surviving legs on failure; took logged queries from 0% to 100% P@1 with deterministic entity resolution (0.4ms) after rejecting embeddings.
- Traced another team's 65% LLM failure rate via Langfuse to an unbounded reasoning loop (p95 1,197 tokens vs. 16,384 runaways); a binomial model predicted 49–66% failure vs. 65% observed.
- Sole SwiftUI engineer; shipped [SportsBreak](#) to the App Store (~6,000 installs, 1,119 peak daily active users).

### Haddee Education

June – September 2025

*Engineering Lead*

*San Diego, CA*

- Led an 8-intern team to take a React, Next.js, FastAPI, and Supabase web app from 0→1 on GCP App Engine; designed database schemas and REST APIs for LangChain resume parsing and AI job matching.

### ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch

September – December 2023

*Full Stack Developer*

*Beijing, China*

- Built a VC-backed startup's LLM mock-interview web app from 0→1 as the sole engineer in React, Next.js, Supabase, and PostgreSQL; owned auth, session state, and database interactions.

## RESEARCH EXPERIENCE

### Control & Learning Group — Carnegie Mellon University

February – June 2026

*Graduate Researcher | Advisor: Prof. Yorie Nakahira*

- Developed an Online Adaptive Probabilistic Safety Certificate (PSC) with PhD researchers for stochastic systems; outperformed Adaptive MPC in 3-DOF vehicle simulations under environmental uncertainty.
- Implemented Bayesian estimators for real-time road-friction adaptation (LuGre tire models) and mapped natural-language instructions to formal safety specs with multi-turn LLMs.

## SKILLS

|               |                                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Languages     | Python, Go, Java, C, SQL, TypeScript, Swift, JavaScript, Bash                     |
| Cloud & Infra | Linux, Kubernetes (EKS), Docker, Helm, AWS, GCP, Jenkins, GitHub Actions, CI/CD   |
| Data & ML     | MongoDB, PostgreSQL, MySQL, PyTorch, NumPy, CUDA, RAG, hybrid retrieval, Langfuse |
| Frameworks    | FastAPI, Pydantic, Motor / async MongoDB, SwiftUI, React, Next.js, LangChain      |
| Practices     | REST APIs, Unit Testing, pytest, pyright, ruff, TDD, Git, Agile/Scrum             |

## PROJECTS

- my-av:** Built a PyTorch CNN-GRU that predicts steering and speed from 20-frame (1s) video; built a data pipeline over 33 hours of Comma2k19 and ran streaming inference on local CUDA and GCP Vertex AI.
- Wisconsin Autonomous:** Integrated wheel-speed encoders on the competition rover and streamed real-time data to the onboard NVIDIA Jetson.